GraphRAG for Biomedical Literature

Knowledge-Graph Augmented Retrieval over Breast Cancer Research

Project Proposal

Domain	Biomedical NLP - Breast Cancer Literature
Corpus	PubMed abstracts & full-text articles
System	GraphRAG - Graph + Vector Hybrid Retrieval
NER approach	scispaCy + BiomedBERT + custom patterns
Graph store	CSV triples (extensible to Neo4j)
Vector store	FAISS IndexFlatIP, BGE-base embeddings
Status	Proposal - awaiting approval

Contents

1. Executive Summary

Biomedical research on breast cancer is vast, fast-moving, and spread across thousands of publications. Researchers and clinicians need to quickly answer questions like: What therapies target breast cancer stem cells? Which biomarkers predict resistance to treatment? What nanoparticle systems have been tested for drug delivery?

This proposal describes a Graph-augmented Retrieval system (GraphRAG) adapted from a working document-QA prototype to operate over biomedical literature. The system combines two retrieval strategies: dense semantic search over encoded text chunks, and structured traversal of a biomedical knowledge graph extracted from the same corpus. Together they return high-precision, citation-backed answers without relying on any external database or paid API.

The system is grounded in the sample text provided, which covers breast cancer heterogeneity, breast cancer stem cells (BCSCs), resistance mechanisms, antibody-drug conjugates (ADCs), nanoparticle delivery systems, and prognostic biomarkers including Oncotype DX, MammaPrint, and uPA/PAI-1.

2. Problem Statement

2.1 The information challenge in breast cancer research

Breast cancer is a heterogeneous disease involving genetic, environmental, and cellular factors. Literature on its treatment spans multiple disciplines: oncology, pharmacology, nanotechnology, molecular biology, and clinical diagnostics. Key challenges for researchers and analysts include:

- Thousands of new articles published per year on PubMed alone.
- Relationships between entities - drugs, genes, cell types, biomarkers - are buried in unstructured text.
- Standard keyword search returns documents but not answers. A query for 'BCSCs resistance' returns papers that mention both words, not papers that explain the mechanism.
- Commercial solutions (e.g. IBM Watson for Oncology) are expensive, opaque, and not customisable to internal research corpora.

2.2 What is missing

There is no lightweight, in-house system that can:

1. Accept a natural-language question about breast cancer treatment or biology.
2. Retrieve the most semantically relevant passages from a curated literature corpus.
3. Surface the structured relationships between the entities mentioned in those passages.
4. Return results with full citation traceability - paper title, author, and section.

3. Proposed Solution - Biomedical GraphRAG

Same two-pipeline architecture as the Xerox manual system - adapted for biomedical NLP

The system operates in two phases. Phase 1 (offline indexing) reads the literature corpus and builds a knowledge graph plus a vector index. Phase 2 (online query) takes a natural-language question and retrieves relevant passages and graph facts simultaneously.

3.1 How it differs from the Xerox manual system

Aspect	Xerox manual system	Biomedical system (proposed)
Corpus	Single printer manual, ~100 pages	PubMed abstracts + full-text articles, thousands of papers
NER method	Compiled regex patterns, 8 types	scispaCy + BiomedBERT, 8 biomedical types
Entity normalisation	None - surface string only	UMLS CUI linking - resolves synonyms to canonical IDs
Embedding model	all-MiniLM-L6-v2 (general)	BGE-base-en or BiomedBERT (domain-tuned)
Chunk metadata	chunk_id, page, text	chunk_id, pmid, title, authors, year, section, text
Relation rules	6 type-pair rules + RELATED_TO fallback	10 biomedical relation rules + ASSOCIATED_WITH fallback
Deduplication	By (source, relation, target, page)	By (CUI_src, relation, CUI_tgt) across all papers

4. Biomedical Entity Type Catalogue

Eight entity types are defined based on the sample corpus. Each type has a set of patterns or a model-based recogniser. All extracted entities are linked to UMLS Concept Unique Identifiers (CUIs) where available, enabling synonym resolution across papers.

Type	Example from sample	Recognition method	UMLS semantic type
DISEASE	Breast cancer, BC	scispaCy + UMLS linking	T047 Disease or Syndrome
CELL_TYPE	Breast cancer stem cells (BCSCs)	scispaCy + custom patterns	T025 Cell
THERAPY	ADCs, nanoparticles, albumin-based NPs	BiomedBERT NER	T061 Therapeutic Procedure
DRUG	Antibody-drug conjugate payload	scispaCy + DrugBank patterns	T121 Pharmacologic Substance
GENE_PROTEIN	uPA, PAI-1	scispaCy + HGNC patterns	T116/T123 Gene/Protein
BIOMARKER	Prognostic biomarker, predictive biomarker	Custom regex + context rules	T201 Clinical Attribute
ASSAY_TEST	Oncotype DX, MammaPrint, uPA/PAI-1 test	Custom patterns + PubMed MeSH	T059 Laboratory Procedure
MECHANISM	Resistance to therapies, aggressiveness	BiomedBERT + relation context	T169 Functional Concept

4.1 Why UMLS linking matters

The same concept appears under many names in biomedical literature. Without normalisation, 'breast cancer', 'BC', 'mammary carcinoma', and 'breast carcinoma' are treated as four different entities. UMLS CUI linking maps all four to the same canonical concept (C0006142), so relationships extracted from different papers can be merged correctly.

```
# Example: synonym resolution via scispaCy + UMLS
raw_text   = 'BC is a heterogeneous disease'
entity     = 'BC'
cui        = linker.resolve('BC')   -> 'C0006142'
canonical  = 'Breast Neoplasms'     <- UMLS preferred term

# All of these now map to the same graph node:
# 'breast cancer' -> C0006142
# 'BC'            -> C0006142
# 'mammary tumor' -> C0006142
```

5. Biomedical Relation Type Catalogue

Ten specific relation types are defined for the biomedical domain, replacing the six used in the Xerox system. All are inferred from the type combination of the source and target entity, with ASSOCIATED_WITH as the fallback for unrecognised pairs.

Relation	Source type	Target type	Example triple
TREATS	THERAPY / DRUG	DISEASE	ADC --TREATS--> Breast Cancer
TARGETS	THERAPY / DRUG	CELL_TYPE	BCSCs-therapy --TARGETS--> BCSCs
DRIVES	CELL_TYPE	MECHANISM	BCSCs --DRIVES--> Aggressiveness
CAUSES_RESISTANCE	MECHANISM	THERAPY	Resistance --CAUSES_RESISTANCE--> Chemotherapy
DELIVERED_VIA	DRUG	THERAPY	Drug --DELIVERED_VIA--> Albumin NP
INDICATES	BIOMARKER	DISEASE	Prognostic BM --INDICATES--> BC outcome
MEASURES	ASSAY_TEST	BIOMARKER	Oncotype DX --MEASURES--> Prognostic BM
EXPRESSED_BY	GENE_PROTEIN	CELL_TYPE	uPA --EXPRESSED_BY--> Tumour Cell
INVOLVED_IN	GENE_PROTEIN	MECHANISM	PAI-1 --INVOLVED_IN--> Resistance
INFLUENCES	MECHANISM	DISEASE	Genetic factors --INFLUENCES--> BC
ASSOCIATED_WITH	Any	Any	Fallback - co-occurrence without specific rule

The ASSOCIATED_WITH fallback is retained but explicitly labelled so analysts know it carries lower confidence than the 10 specific types.

6. Pipeline A - Offline Indexing

Trigger: one-time run per corpus update | Script: bio_indexer.py

6.1 Corpus ingestion

Each PubMed article is stored as a structured record. The chunk metadata is richer than the Xerox system to enable full citation traceability.

```
INPUT record per article:
{
  'pmid'    : '12345678',
  'title'   : 'Breast cancer stem cells and resistance...',
  'authors' : ['Smith J', 'Patel R'],
  'year'    : 2023,
  'section' : 'Abstract | Introduction | Methods | Results | Discussion',
  'text'    : 'During the past recent years...'
}
```

6.2 Text chunking

Same sliding-window algorithm as the Xerox system. Parameters are tightened for biomedical text because abstracts are shorter and denser than manual pages.

Parameter	Value	Rationale
`CHUNK_SIZE`	800 chars	Abstracts average 250 words; 800 chars captures ~1 paragraph
`CHUNK_OVERLAP`	100 chars	Prevents entity span splits at sentence boundaries
`stride`	700 chars	CHUNK_SIZE minus CHUNK_OVERLAP

6.3 Biomedical NER - two-stage approach

The Xerox system used compiled regex patterns which work well for structured technical text (model numbers, menu names). Biomedical text requires model-based recognition because entity boundaries are irregular and context-dependent.

Stage 1 - scispaCy pipeline

```
import spacy
from scispacy.linking import EntityLinker

nlp = spacy.load('en_core_sci_lg')
nlp.add_pipe('scispacy_linker',
             config={'resolve_abbreviations': True,
                     'linker_name': 'umls'})

doc = nlp(chunk_text)
for ent in doc.ents:
```

```
    label = ent.label_          # e.g. 'DISEASE', 'GENE_OR_GENE_PRODUCT'
    text  = ent.text            # surface form
    cui   = ent._.kb_ents[0][0] # UMLS CUI (top candidate)
    score = ent._.kb_ents[0][1] # linking confidence
```

Stage 2 - custom pattern layer

A small set of regex patterns is layered on top for entity types scispaCy misses, such as assay names and nanoparticle subtypes.

```
BIOMEDICAL_PATTERNS = {
  'ASSAY_TEST' : [r'Oncotype\s+DX', r'MammaPrint',
                  r'uPA[/\s]PAI-1', r'Mamm\w+Print'],
  'THERAPY'    : [r'antibody[- ]drug\s+conjugate',
                  r'ADC', r'nanoparticle',
                  r'albumin[- ]based', r'lipid[- ]based',
                  r'polymer[- ]based', r'micelle[- ]based'],
  'CELL_TYPE'  : [r'breast\s+cancer\s+stem\s+cells?',
                  r'BCSCs?', r'tumou?r\s+cells?'],
  'MECHANISM'  : [r'resistance\s+to\s+therap',
                  r'aggressiveness', r'heterogeneous?']
}
```

6.4 Knowledge graph construction

Entity pairs are combined at sentence level (not chunk level) to reduce co-occurrence noise - the key fix identified in the Xerox system. infer_relation() uses 10 biomedical-specific rules before falling back to ASSOCIATED_WITH.

```
import nltk

FOR chunk IN chunks:
    FOR sentence IN nltk.sent_tokenize(chunk.text):   # sentence scope
        entities = extract_entities(sentence)
        unique   = deduplicate_by_cui(entities)
        FOR i, src IN enumerate(unique):
            FOR tgt IN unique[i+1:]:
                relation = infer_relation(src.type, tgt.type)
                key = (src.cui, relation, tgt.cui)   # CUI-based dedup
                IF key NOT IN seen: emit triple

WRITES: bio_graph_triples.csv
SCHEMA: src_text, src_cui, src_type, relation,
        tgt_text, tgt_cui, tgt_type,
        pmid, year, section, evidence[:400]
```

6.5 Vector index

A biomedical-domain embedding model is used instead of the general-purpose MiniLM. BGE-base-en-v1.5 or BiomedBERT produce embeddings that cluster biomedical synonyms more tightly.

```
# Option A: BGE (strong general + biomedical performance)
model = SentenceTransformer('BAAI/bge-base-en-v1.5')
```

```
# Option B: BiomedBERT (domain-tuned on PubMed)
model = SentenceTransformer('pritamdeka/BioBERT-mnli-snli-scinli-scitail-mednli-stsb')

embeddings = model.encode(texts,
                 normalize_embeddings=True,
                 show_progress_bar=True)
index = faiss.IndexFlatIP(768)   # dim=768 for BGE/BiomedBERT
index.add(embeddings)

WRITES: bio_manual.faiss
        bio_chunk_metadata.json   # includes pmid, title, year, section
```

6.6 Output artifacts

File	Format	Contents
`bio_chunks.jsonl`	JSONL	pmid, title, year, section, chunk_id, text
`bio_entities.json`	JSON	Entity list with CUI, type, count, pmids[], chunks[]
`bio_graph_triples.csv`	CSV	src_text, src_cui, relation, tgt_text, tgt_cui, pmid, evidence
`bio_manual.faiss`	Binary	IndexFlatIP, dim=768, n=chunk count, float32
`bio_chunk_metadata.json`	JSON	Ordered list: chunk_id, pmid, title, authors, year, section, text

7. Pipeline B - Online Query

Trigger: each research question | Script: bio_query.py | No external API

7.1 vector_search(question, top_k=5)

Identical logic to the Xerox system but using the biomedical embedding model and 768-dim index. Results include pmid and paper title so every passage is immediately citable.

```
q_emb = model.encode([question],
            normalize_embeddings=True).astype('float32')
scores, ids = index.search(q_emb, top_k)

FOR rank, idx IN enumerate(ids[0]):
    chunk = chunks[idx]
    yield {
        'score'   : float(scores[0][rank]),
        'chunk_id': chunk['chunk_id'],
        'pmid'    : chunk['pmid'],
        'title'   : chunk['title'],
        'year'    : chunk['year'],
        'section' : chunk['section'],
        'text'    : chunk['text']
    }
```

7.2 graph_search(question, vector_results, limit=20)

Same scoring logic. Stopword list is extended with common biomedical filler terms. Field-weighted scoring is applied: entity name hits score higher than evidence hits.

```
BIO_STOPWORDS = {'how','what','why','when','the','and',
                 'for','are','can','you','do','is','to',
                 'with','from','that','this','were','has',
                 'have','been','their','which','also'}

FOR triple IN triples:
    score = 0
    IF triple['chunk_id'] IN hit_chunks: score += 5
    FOR term IN clean_terms:
        IF term IN (src_text + tgt_text).lower(): score += 3
        ELIF term IN relation.lower():             score += 2
        ELIF term IN evidence.lower():             score += 1
    IF score > 0: keep(score, triple)

dedup by (src_cui, relation, tgt_cui)   # CUI-level dedup
return top 20
```

7.3 Final output format

Results are printed to stdout with full citation metadata. Each vector result includes the PMID, paper title, publication year, and section, making every returned passage immediately traceable to its source publication.

```
QUESTION
What therapies target breast cancer stem cells?

VECTOR RESULTS
----------------------------------------------------------
PMID: 12345678 | Year: 2023 | Section: Abstract
Title: Breast cancer stem cells and resistance to therapy
Score: 0.731
Breast cancer stem cells (BCSCs) are the main player in
aggressiveness... BCSCs-based therapies target these cells...

GRAPH RESULTS
BCSCs-therapy --TARGETS--> BCSCs        | PMID 12345678
BCSCs --DRIVES--> Aggressiveness         | PMID 12345678
ADC --TREATS--> Breast Cancer            | PMID 12345678
Nanoparticle --TREATS--> Breast Cancer   | PMID 12345678
```

8. Traced Example - Sample Text

Input: the breast cancer review abstract provided

The following section traces exactly what the system would extract from the provided sample paragraph, step by step.

8.1 Input text

"During the past recent years, various therapies emerged in the era of breast cancer. Breast cancer is a heterogeneous disease in which genetic and environmental factors are involved. Breast cancer stem cells (BCSCs) are the main player in the aggressiveness of different tumors... the major obstacle to achieve an effective treatment is resistance to therapies... antibody-drug conjugation systems (ADCs), nanoparticles (albumin-, metal-, lipid-, polymer-, micelle-based nanoparticles), and BCSCs-based therapies... Oncotype DX, MammaPrint, and uPA/PAI-1 are regarded as suitable prognostic and predictive factors."

8.2 Entities extracted

Surface form	Type	UMLS CUI	Canonical name
breast cancer	DISEASE	`C0006142`	Breast Neoplasms
BC	DISEASE	`C0006142`	Breast Neoplasms (resolved abbreviation)
Breast cancer stem cells	CELL_TYPE	`C2986879`	Breast Cancer Stem Cells
BCSCs	CELL_TYPE	`C2986879`	Breast Cancer Stem Cells (abbreviation resolved)
antibody-drug conjugation systems	THERAPY	`C3272898`	Antibody-Drug Conjugates
ADCs	THERAPY	`C3272898`	Antibody-Drug Conjugates (resolved)
albumin-based nanoparticles	THERAPY	`C0021267`	Nanoparticles, Albumin-Based
lipid-based nanoparticles	THERAPY	`C1721048`	Lipid Nanoparticles
Oncotype DX	ASSAY_TEST	`C2986547`	Oncotype DX Assay
MammaPrint	ASSAY_TEST	`C2350535`	MammaPrint Gene Signature
uPA	GENE_PROTEIN	`C0041350`	Urokinase-Type Plasminogen Activator
PAI-1	GENE_PROTEIN	`C0032588`	Plasminogen Activator Inhibitor 1
resistance to therapies	MECHANIS	`C0013203`	Drug Resistance

	M		
aggressiveness	MECHANIS M	C1511890	Tumor Aggressiveness
prognostic biomarker	BIOMARKE R	C0005516	Biological Markers, Prognostic
predictive biomarker	BIOMARKE R	C0005516	Biological Markers, Predictive

8.3 Triples extracted (sentence-level)

Source entity	Relation	Target entity	Confidence
BCSCs	DRIVES	Aggressiveness	HIGH
BCSCs	DRIVES	Drug Resistance	HIGH
ADC	TREATS	Breast Cancer	HIGH
Albumin NP	TREATS	Breast Cancer	HIGH
Lipid NP	TREATS	Breast Cancer	HIGH
BCSCs-based therapy	TARGETS	BCSCs	HIGH
Oncotype DX	MEASURES	Prognostic Biomarker	HIGH
MammaPrint	MEASURES	Prognostic Biomarker	HIGH
uPA/PAI-1	MEASURES	Prognostic Biomarker	HIGH
Drug Resistance	CAUSES_RESISTANCE	Effective Treatment	HIGH
Genetic factors	INFLUENCES	Breast Cancer	MED
Environmental factors	INFLUENCES	Breast Cancer	MED

9. Technology Stack

Layer	Library / Tool	Purpose
NER	`scispaCy en_core_sci_lg`	Biomedical entity recognition + UMLS linking
NER	`BiomedBERT`	Context-sensitive recognition for complex entity spans
Embeddings	`BAAI/bge-base-en-v1.5`	768-dim domain-aware sentence embeddings
Vector index	`FAISS IndexFlatIP`	Exact cosine search, suitable up to ~500K chunks
Sentence split	`nltk.sent_tokenize`	Sentence-level entity pairing to reduce co-occurrence noise
UMLS linking	`scispaCy EntityLinker`	CUI resolution, abbreviation expansion, synonym merging
Corpus source	`PubMed / PMC API`	Bulk article download; abstracts free, full text via PMC
Corpus parsing	`lxml / Biopython Entrez`	PubMed XML parsing into structured records
Graph store	`CSV (phase 1) / Neo4j`	CSV for prototype; Neo4j AuraDB for production scale
Orchestration	`Python 3.11`	Same pipeline structure as existing Xerox system

10. Key Design Decisions

10.1 Sentence-level entity pairing

The Xerox system paired entities at chunk level, producing spurious triples from unrelated concepts that happened to appear in the same 1,200-char window. For biomedical text this is a larger problem because abstracts are dense and contain many entity types per paragraph.

Decision: restrict entity pairing to sentence boundaries using nltk.sent_tokenize(). Two entities must appear in the same sentence to form a triple. This reduces triple count but significantly improves precision.

10.2 CUI-based deduplication

The Xerox system deduplicated by surface string, causing 'Network' and 'network' to produce duplicate triples. In biomedical text the problem is far worse: a single concept may appear as 'breast cancer', 'BC', 'mammary carcinoma', 'invasive ductal carcinoma', and dozens of other surface forms across papers.

Decision: deduplicate triples by (src_cui, relation, tgt_cui). Two triples from different papers that refer to the same conceptual relationship are merged into one, with a pmids[] list recording all source papers.

10.3 Domain-tuned embeddings

General-purpose models like all-MiniLM-L6-v2 treat 'BC' and 'breast cancer' as different tokens with different embedding positions. Biomedical models trained on PubMed cluster these correctly because they have seen the co-occurrence pattern millions of times.

Decision: use BGE-base-en-v1.5 or BiomedBERT. Both produce 768-dim embeddings. The FAISS index dimension changes from 384 to 768 accordingly.

10.4 Citation metadata per chunk

The Xerox system stored only page number and text in chunk metadata. For literature review use, a result without a citation is not useful.

Decision: store pmid, title, authors, year, and section in every chunk record so every vector result is immediately citable without any additional lookup.

11. Phased Delivery Plan

Phase	Deliverable	Activities	Effort
1	Corpus + chunking	PubMed API download, XML parsing, sliding-window chunker, chunk_metadata with citation fields	1 week
2	NER + UMLS linking	scispaCy pipeline, custom pattern layer, CUI resolution, abbreviation expansion, entity store	1-2 weeks
3	Graph construction	Sentence-level entity pairing, 10 relation rules, CUI dedup, graph_triples CSV	1 week
4	Vector index	BGE/BiomedBERT encoding, FAISS IndexFlatIP build, index validation	3 days
5	Query engine	vector_search + graph_search with biomedical stopwords + field-weighted scoring	3 days
6	Evaluation	Batch test on 20 representative questions; precision / recall assessment on graph triples	1 week
Total	**Working prototype**	End-to-end biomedical GraphRAG - indexing + query - no external API	**5-7 weeks**

12. Expected Outputs

12.1 Answerable question types

The completed system will be able to retrieve high-quality context for the following research question categories:

- Therapy questions: Which therapies treat breast cancer stem cells? What nanoparticle types have been studied for BC drug delivery?
- Mechanism questions: What drives resistance to therapy in BC? How does BCSC aggressiveness relate to treatment failure?
- Biomarker questions: Which assays measure prognostic biomarkers? What is the role of uPA/PAI-1 in BC prognosis?
- Gene/protein questions: What mechanisms involve PAI-1? Which proteins are expressed by breast cancer stem cells?
- Comparative questions: How does MammaPrint compare to Oncotype DX as a predictive factor?

12.2 Sample expected output

For the question 'What drives aggressiveness in breast cancer?', the system would return:

```
VECTOR RESULTS
PMID 12345678 | 2023 | Abstract | Score: 0.81
Breast cancer stem cells (BCSCs) are the main player in
the aggressiveness of different tumors...

GRAPH RESULTS
BCSCs --DRIVES--> Aggressiveness          | PMID 12345678
BCSCs --DRIVES--> Drug Resistance         | PMID 12345678
Genetic factors --INFLUENCES--> Breast Cancer | PMID 12345678
```

13. Glossary

Term	Definition
GraphRAG	Graph-augmented Retrieval system. Combines knowledge-graph traversal with dense vector search.
UMLS	Unified Medical Language System. A large biomedical ontology that maps synonymous terms to Concept Unique Identifiers (CUIs).
CUI	Concept Unique Identifier. A stable UMLS ID for a biomedical concept, e.g. C0006142 = Breast Neoplasms.
scispaCy	A Python NLP library built on spaCy, trained on biomedical text. Includes an EntityLinker component for UMLS resolution.
BiomedBERT	A BERT model pre-trained on PubMed abstracts and full-text articles. Produces domain-aware embeddings for biomedical text.
BGE	BAAI General Embeddings. A family of sentence embedding models with strong performance on biomedical retrieval benchmarks.
BCSC	Breast Cancer Stem Cell. A subpopulation of cancer cells with self-renewal capacity, linked to tumour aggressiveness and therapy resistance.
ADC	Antibody-Drug Conjugate. A targeted therapy that attaches a cytotoxic drug to an antibody directed at a cancer-specific antigen.
Oncotype DX	A genomic assay that measures the expression of 21 genes in breast tumour tissue to predict recurrence risk and chemotherapy benefit.
MammaPrint	A 70-gene expression signature assay used as a prognostic and predictive biomarker in early-stage breast cancer.
uPA / PAI-1	Urokinase Plasminogen Activator and its inhibitor. Protein biomarkers used to predict chemotherapy benefit in node-negative BC.
FAISS	Facebook AI Similarity Search. A library for efficient dense vector similarity search. IndexFlatIP = exact inner product search.
Co-occurrence	Two entities appearing in the same text unit (sentence or chunk). Sentence-level co-occurrence has higher precision than chunk-level.
Triple	A (subject, predicate, object) fact, e.g. BCSCs --DRIVES--> Aggressiveness. The atomic unit of a knowledge graph.